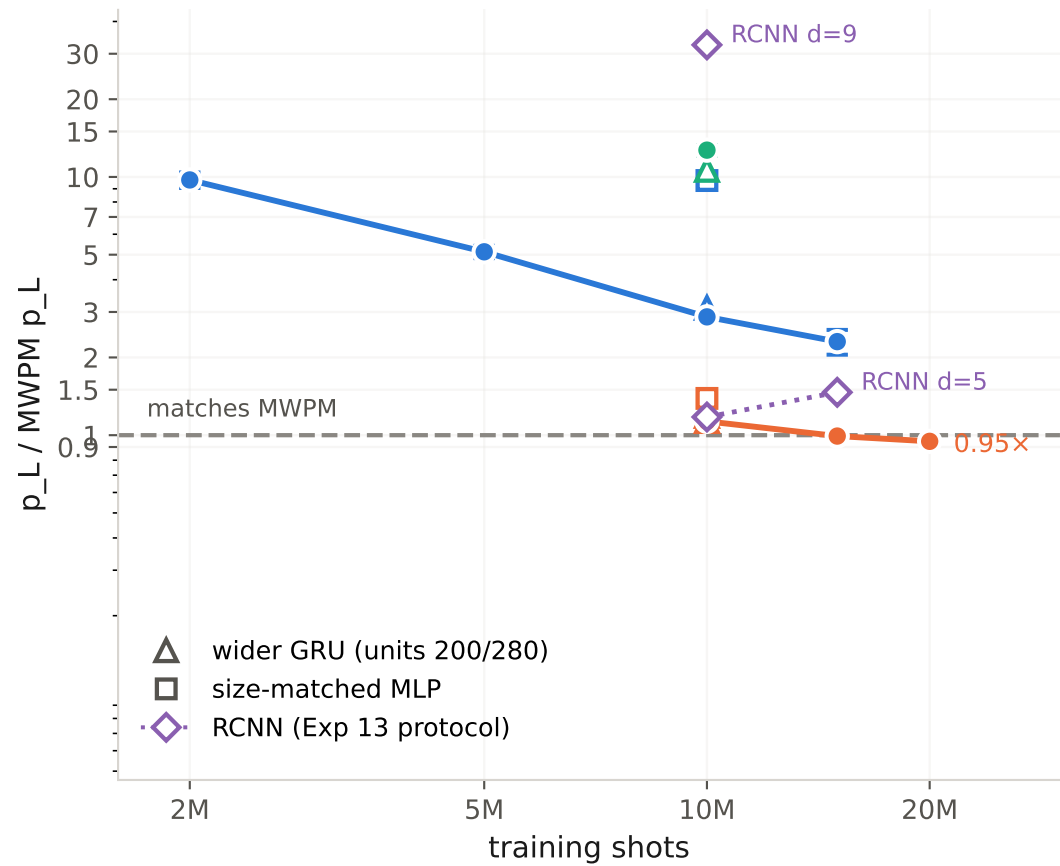
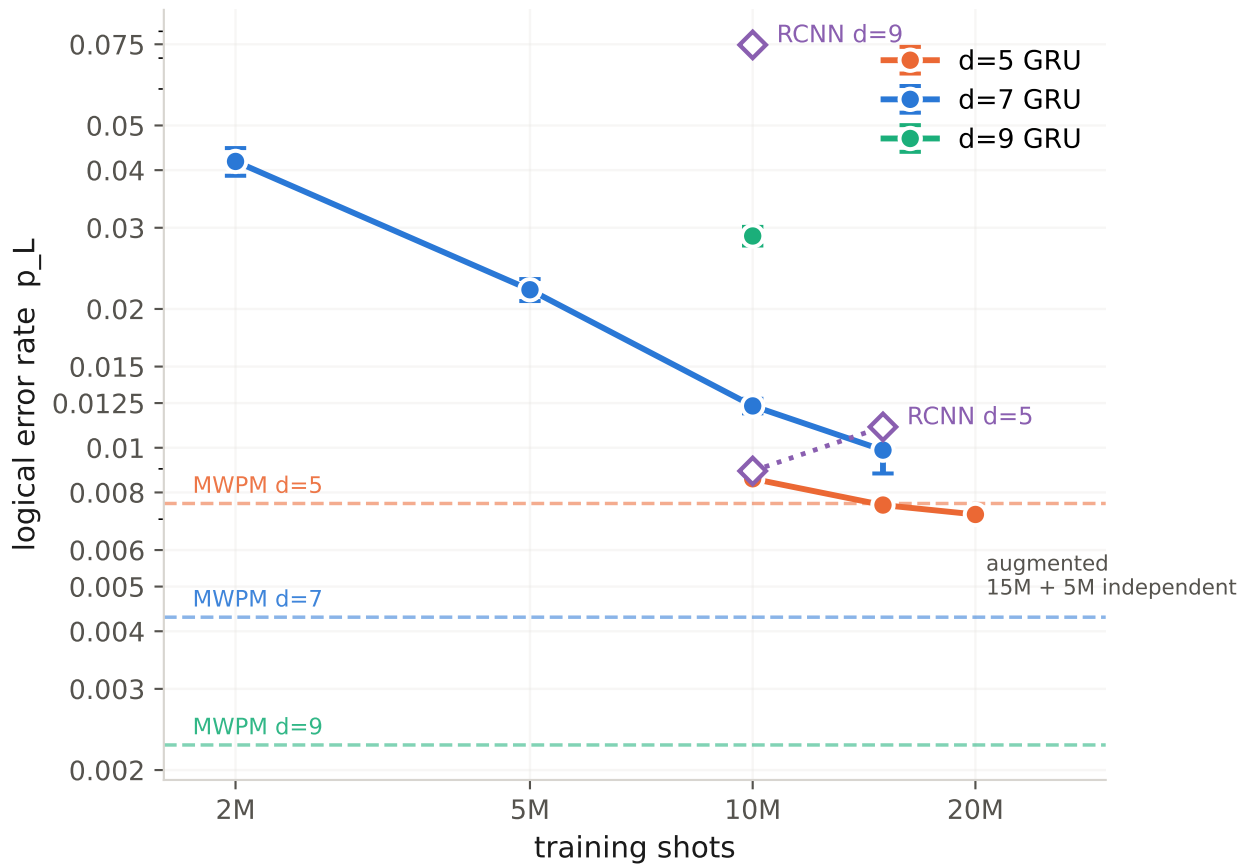


Experiment 16 — training data is what closes the gap to MWPM, and the amount needed grows with distance



Left: absolute error rate against training shots; dashed lines are each distance's MWPM. Right: the same runs relative to their own MWPM. GRU points are units=140, batch 10,000, 200 fixed epochs, minimum-validation-loss checkpoint, $p=0.004$, scored on [15.2M, 17.0M) with MWPM decoded on those shots; error bars are the standard deviation over three seeds. Open triangles and squares are single-seed probes at 10M. The d=5 20M point is an augmented set (15M primary + 5M independently generated), not a prefix. RCNN is dotted and follows Experiment 13's protocol -- batch 5,000, epoch-indexed LR schedule with early stopping, one seed, EAF pools -- so it shares the axes but not the recipe; its d=5 pair gets worse with more data.